

Minimum Subspace Trades and Relation Lattices in Three Binary Designs

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Abstract

We study the point-block relation modules of three fixed collections of four-dimensional subspaces of $\mathbb{F}_2^8$, each invariant under the same group of order 204. The minimum point trades are the six-block opposite-regulus trades. We determine their exact censuses, prove that they span the full rational relation module in each of the three configurations, and identify the complete degree-three layer of the associated incidence toric ideals. The minimum trades are locally saturated at 2, 3, 17, 137, 219 097, and 3 288 036 131. At 137, a length-one defect in a row presentation selected at the control prime 409 disappears when all minimum-circuit orbits are imposed. The same full-circuit restoration occurs at 219 097 and 3 288 036 131. Exact maximal-minor screens, independently replayed at 128 localizations over 219 097 and 80 localizations over 3 288 036 131, close the other five large component maps at those primes. Beyond these finite screens, two exact characteristic-zero maximal minors prove that the complete $\rho/\mathcal{B}_1$ component map is surjective over $\mathbb{Z}[\zeta_{17}, 1/(2 \cdot 3 \cdot 17)]$. The corresponding global gates for the other five large maps and the global Smith groups remain open.

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1 Introduction

Let $\mathcal{P}$ be the set of 255 points of $\text{PG}(7, 2)$. We fix three block collections $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, each consisting of 66 929 four-dimensional vector subspaces of $\mathbb{F}_2^8$. They arise as three constituents in the locked presentation studied in the companion construction paper; their construction lineage is described in [9]. The present paper starts from that finite triple and studies a different layer: relations among its point-incidence columns.

Write

$$A_i : \mathbb{Z}[\mathcal{B}_i] \longrightarrow \mathbb{Z}[\mathcal{P}], \quad B \longmapsto \sum_{x \in \text{PG}(B)} x.$$

Every A_i has rank 255. We compare the rational relation module $K_i = \ker(A_i \otimes \mathbb{Q})$ with the submodule generated by the smallest possible point trades. Those trades have three blocks in each leg and come from the two reguli in a quotient $\text{PG}(3, 2)$.

The three configurations have the same parameters and a common symmetry group

$$\Gamma \cong (C_{17} \rtimes C_4) \times C_3, \quad |\Gamma| = 204,$$

but their relation modules are not isomorphic. This is already visible in every rational irreducible type. Despite that separation, the same local six-block geometry generates all rational relations in all three cases.

Theorem 1.1 (Main theorem). *For the locked triple $(\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3)$, the following hold.*

- (i) *The rational $\mathbb{Q}\Gamma$ -modules K_1, K_2, K_3 are pairwise nonisomorphic, and their multiplicities differ in each of the eight rational irreducible types.*
- (ii) *The $\mathbb{Q}$ -span of the minimum opposite-regulus trades is K_i for every i .*
- (iii) *The associated incidence toric ideal has no nonzero element of degree below three. Its primitive cubics are precisely the minimum opposite-regulus trades; every such cubic is indispensable.*
- (iv) *If L_i is the integral lattice generated by the minimum trades and*

$$\Theta_i = \ker_{\mathbb{Z}}(A_i)/L_i,$$

then Θ_i is finite and has zero 2-, 3-, and 17-primary parts.

- (v) *The complete primary integral quotients vanish at all three selected good primes:*

$$\Theta_i[p^\infty] = 0 \quad (i = 1, 2, 3, \ p \in \{137, 219\,097, 3\,288\,036\,131\}).$$

The apparent selected-row $\rho/\mathcal{B}_1$ defects at these primes disappear when the complete minimum-circuit equation maps are imposed.

- (vi) *For the complete $\rho/\mathcal{B}_1$ minimum-circuit equation map over $R = \mathbb{Z}[\zeta_{17}]$,*

$$\text{coker}(F_{\rho,1}^{\text{full}}) \otimes_R R \left[\frac{1}{2 \cdot 3 \cdot 17} \right] = 0.$$

Hence this component has no prime-ideal support away from the rational primes 2, 3, and 17.

The finite-prime assertions at 137, 219 097, and 3 288 036 131 are complete for the frozen triple. The last item is stronger but narrower in a different direction: it is global away from 2, 3, and 17, yet applies only to the complete $\rho/\mathcal{B}_1$ component. The corresponding global gates for the other five large maps remain open, so no support theorem or global Smith classification for the complete groups Θ_i is asserted. Likewise, our toric conclusion concerns the first nonzero graded layer; it does not assert that all higher-degree moves are generated by cubics.

The proof has three interfaces. Geometry identifies and counts the minimum circuits. Semisimple representation theory converts selected circuits into exact component-rank problems and proves rational generation. Integral localization first proves saturation at the tested primes and exposes why control-prime row selection cannot certify a good-prime defect; an exact cyclotomic two-minor Fitting certificate then closes the complete $\rho/\mathcal{B}_1$ component away from 2, 3, and 17. All numerical claims are accompanied by exact finite certificates; the certificate contract is described in Section A.

Organization. Section 2 fixes notation and the rational characters. Section 3 decomposes the relation modules. Section 4 proves the local circuit statement and gives the exhaustive census. Section 5 proves rational generation, and Section 6 determines the cubic toric layer. Section 7 treats the primes 2, 3, 17. Section 8 proves the complete primary classifications at 137, 219 097, and 3 288 036 131. Section 9 gives the global localized theorem for the single complete $\rho/\mathcal{B}_1$ component. Open integral and higher-degree questions are separated in Section 10.

2 The locked triple and its common symmetry

All subspace dimensions in this paper are vector dimensions. Thus a block $B \in \mathcal{B}_i$ is a four-dimensional subspace of $\mathbb{F}_2^8$ and contains 15 projective points. The free block and point lattices are

$$M_i^{\mathbb{Z}} = \mathbb{Z}[\mathcal{B}_i], \quad P^{\mathbb{Z}} = \mathbb{Z}[\mathcal{P}].$$

We use superscripts $\mathbb{Q}$ and k for scalar extension. The integral and rational relation modules are

$$K_i^{\mathbb{Z}} = \ker(A_i : M_i^{\mathbb{Z}} \rightarrow P^{\mathbb{Z}}), \quad K_i = K_i^{\mathbb{Z}} \otimes \mathbb{Q}.$$

Since A_i has full row rank,

$$\dim_{\mathbb{Q}} K_i = 66\,929 - 255 = 66\,674.$$

The common group is generated by elements a, b, c satisfying

$$a^{17} = b^4 = c^3 = 1, \quad bab^{-1} = a^4, \quad [a, c] = [b, c] = 1.$$

Set $F = \langle a, b \rangle \cong C_{17} \rtimes C_4$. The action of $\Gamma = F \times \langle c \rangle$ preserves every block collection and the point set, so every incidence map A_i is Γ -equivariant.

2.1 The eight rational types

The rational irreducibles can be constructed without choosing a matrix model. The quotient $F/C_{17} \cong C_4$ has the three rational types $\mathbf{1}, \varepsilon, \iota$, of dimensions 1, 1, 2. Here ε is the order-two character and ι is the rational form of the pair of order-four characters. The group C_3 has rational types $\mathbf{1}, \omega$, where ω is the two-dimensional rational form of the two nontrivial complex characters.

For a nontrivial complex character χ of C_{17} , the C_4 action has an orbit of size four, and $\text{Ind}_{C_{17}}^F \chi$ has complex degree four. The four such induced characters form one Galois orbit. Their rational form is denoted ρ and has dimension 16. Tensoring with the nontrivial C_3 -orbit gives $\rho\omega$, of dimension 32.

Thus the rational types and dimensions are

V	$\mathbf{1}$	ε	ι	ω	$\varepsilon\omega$	$\iota\omega$	ρ	$\rho\omega$
$\dim_{\mathbb{Q}} V$	1	1	2	2	2	4	16	32.

For completeness, over $\mathbb{C}$ the first six rational types account for all 12 linear characters of Γ . The type ρ accounts for four degree-four characters and $\rho\omega$ for the remaining eight. Consequently

$$12 \cdot 1^2 + 12 \cdot 4^2 = 204,$$

so the list exhausts the complex irreducibles and hence the rational types.

Proposition 2.1 (Point module). *The rational point permutation module is*

$$\mathbb{Q}[\mathcal{P}] \cong 3\mathbf{1} \oplus 2\varepsilon \oplus 3\omega \oplus 2\varepsilon\omega \oplus 5\rho \oplus 5\rho\omega.$$

In particular, neither ι nor $\iota\omega$ occurs.

Proof. For each of the 24 conjugacy classes, the permutation character equals the number of fixed projective points. Exact inner products with the Galois sums constructed above give the displayed multiplicities. Their dimension-weighted sum is

$$3 + 2 + 6 + 4 + 80 + 160 = 255,$$

which checks the full point dimension. The class values and inner products are included in the electronic representation certificate described in Section A. $\square$

Remark 2.2. *Only the locked finite triple is considered here. No family classification is inferred from its common group or its character table.*

3 Rational relation modules

Because $A_i : M_i^{\mathbb{Q}} \rightarrow P^{\mathbb{Q}}$ is a surjective $\mathbb{Q}\Gamma$ -map, semisimplicity gives

$$[K_i] = [M_i^{\mathbb{Q}}] - [P^{\mathbb{Q}}]$$

in the rational representation ring. Exact character inner products therefore determine K_i from the block and point permutation characters.

Proposition 3.1 (Complete relation-module table). *The multiplicities of the eight rational types in K_1, K_2, K_3 are as follows.*

V	K_1	K_2	K_3	$\dim_{\mathbb{Q}} V$
1	343	354	355	1
ε	341	352	351	1
ι	314	304	302	2
ω	339	349	353	2
$\varepsilon\omega$	338	347	349	2
$\iota\omega$	314	304	302	4
ρ	1312	1314	1310	16
$\rho\omega$	1305	1304	1306	32
<i>weighted total</i>	66 674	66 674	66 674	

In particular, K_1, K_2, K_3 are pairwise nonisomorphic as $\mathbb{Q}\Gamma$ -modules.

Proof. Subtract the point multiplicities in theorem 2.1 from the three exact block permutation characters. The resulting table is the one displayed. For example, the two large-type contributions are

$$16(1312, 1314, 1310) \quad \text{for } \rho, \quad 32(1305, 1304, 1306) \quad \text{for } \rho\omega.$$

Summing every row with its rational dimension gives 66 674 in each column, agreeing with the incidence rank.

The three entries in every row are not all equal; more strongly, each pair of columns differs in at least one multiplicity, and inspection shows that the three configurations are separated in every rational type. Uniqueness of semisimple multiplicities proves pairwise nonisomorphism. $\square$

Remark 3.2 (What the table separates). *The trivial multiplicities 343, 354, 355 are the dimensions of the invariant relation spaces and can also be recovered from block-orbit counts. The structural conclusion is the simultaneous decomposition of the full 66 674-dimensional kernels. It should not be confused with equality of incidence images: all three maps have the same full rational image $P^{\mathbb{Q}}$, while their kernels are inequivalent.*

4 Minimum subspace trades

A point trade in $\mathcal{B}_i$ is an integral relation written with disjoint positive and negative supports,

$$T = T^+ - T^- = \sum_{B \in T^+} B - \sum_{B \in T^-} B \in K_i^{\mathbb{Z}}.$$

Its volume is $|T^+| = |T^-|$, and its total support is $|T^+| + |T^-|$. This distinction matters here: the minimum $T_2(1, 4, 8)$ trades have volume 3 and total support 6.

4.1 The opposite-regulus model

Choose vector subspaces $Z < U < \mathbb{F}_2^8$ with $\dim Z = 2$ and $\dim U = 6$. A four-space B satisfying $Z < B < U$ projects to a projective line B/Z in $\text{PG}(U/Z) \cong \text{PG}(3, 2)$. Three pairwise skew lines form one regulus of the hyperbolic quadric $Q^+(3, 2)$; the three lines meeting all of them form its opposite regulus. Lifting the two reguli gives two triples of four-spaces

$$\{B_1, B_2, B_3\}, \quad \{B'_1, B'_2, B'_3\},$$

with the same point-incidence sum.

Lemma 4.1 (Primitive regulus circuit). *Let N be the 255×6 point-incidence matrix of the six lifted blocks, ordered by the two reguli. Then*

$$\text{rank}_{\mathbb{Q}} N = 5, \quad \ker_{\mathbb{Z}} N = \mathbb{Z}(1, 1, 1, -1, -1, -1).$$

Every five columns are independent, and the nonzero Smith factors of N are $1, 1, 1, 1, 1$. Hence the displayed trade is a primitive unimodular matroid circuit and a Graver element.

Proof. In the quotient $\text{PG}(3, 2)$, each point of the hyperbolic quadric lies on one line from each regulus, giving the displayed kernel vector. Direct row reduction of the 15-point quotient incidence matrix has rank five, and deleting any one line leaves five independent columns. Lifting through $Z < U$ repeats rows but introduces no new column relation. The active Smith form is $\text{diag}(1, 1, 1, 1, 1)$, so the one-dimensional integral kernel is primitive and generated by the sign vector above. A primitive support-minimal relation is a Graver element. $\square$

Krotov's minimum-trade theorem applies with

$$0 \leq t = 1 < k = 4 < v - t = 7.$$

Its support bound specializes to $\prod_{j=0}^1 (1 + 2^j) = 6$, hence volume three, and its $t = 1$ classification identifies every trade attaining this bound with the opposite-regulus construction above [7]. We use that classification for exhaustiveness, while theorem 4.1 supplies the integral circuit statement.

4.2 Exact census

Let $\mathcal{C}_i$ be the set of unoriented minimum trades wholly contained in $\mathcal{B}_i$. Exhaustive enumeration first chooses (Z, U) , constructs the two reguli in U/Z , and then tests membership of all six lifts in $\mathcal{B}_i$. Canonical sorting removes repeated presentations, and the common group then partitions the result into orbits.

Proposition 4.2 (Minimum-trade census). *The numbers of minimum trades and their Γ -orbit counts are*

	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$ \mathcal{C}_i $	2 271 676	2 213 026	2 216 868
$ \mathcal{C}_i/\Gamma $	11 274	11 001	11 033.

Every trade in the census has coefficient gcd one and active Smith form 1^5 . Thus every local arithmetic multiplicity is one.

Proof. Krotov's $t = 1$ classification reduces the search to the (Z, U) -regulus templates, so the enumeration is exhaustive rather than a bounded-support sample. Three independent passes—template generation, direct six-block membership, and orbit closure—give the displayed totals. For each accepted template, theorem 4.1 gives the coefficient gcd and Smith form. The frozen counts, orbit representatives, and replay hashes form the census certificate of Section A. $\square$

Remark 4.3. *The configurations are separated by how many minimum circuits they contain and how those circuits are organized into group orbits. They are not separated by a different six-block local arithmetic type. Nothing in this section classifies minimum $T_2(2, 4, 8)$ trades; for that parameter the same support formula gives 30, or volume 15.*

5 Full rational generation

Let $C_i \subseteq K_i$ be the rational span of all trades in $\mathcal{C}_i$. We prove $C_i = K_i$ by testing the quotient in every irreducible component. This section states both the abstract argument and the exact finite certificate to which it is applied.

5.1 Component equation matrices

Let k be a splitting field for Γ whose characteristic does not divide 204, and let V be an absolutely irreducible $k\Gamma$ -module. Choose one representative in each block orbit and write its stabilizer as H_j . Frobenius reciprocity gives

$$\mathrm{Hom}_\Gamma(k[\mathcal{B}_i], V) \cong \bigoplus_j V^{H_j}.$$

The analogous point-orbit decomposition gives $\mathrm{Hom}_\Gamma(k[\mathcal{P}], V)$. Put

$$m_{B_i}(V) = \dim_k \mathrm{Hom}_\Gamma(k[\mathcal{B}_i], V), \quad m_P(V) = \dim_k \mathrm{Hom}_\Gamma(k[\mathcal{P}], V).$$

Choose minimum-trade orbit seeds $c_1, \dots, c_s$. Evaluation on these seeds defines the component equation map

$$E_{i,V} : \mathrm{Hom}_\Gamma(k[\mathcal{B}_i], V) \longrightarrow V^s, \quad \phi \longmapsto (\phi(c_1), \dots, \phi(c_s)).$$

Once bases of all fixed spaces have been frozen, this is an ordinary matrix over k . Killing one seed kills its entire Γ -orbit, because $\phi(gc) = g\phi(c)$.

Every map $k[\mathcal{P}] \rightarrow V$, pulled back along incidence, kills all circuits. Since the incidence map is surjective, this pullback is injective. Therefore

$$\mathrm{rank} E_{i,V} \leq m_{B_i}(V) - m_P(V).$$

Lemma 5.1 (Component rank criterion). *If*

$$\mathrm{rank} E_{i,V} = m_{B_i}(V) - m_P(V)$$

for every absolutely irreducible V , then $C_i = K_i$.

Proof. The kernel of $E_{i,V}$ is $\mathrm{Hom}_\Gamma(k[\mathcal{B}_i]/C_i, V)$. Equality in the rank bound says that this kernel has dimension $m_P(V)$, exactly the dimension of the incidence pullback. Hence

$$\mathrm{Hom}_\Gamma(k[\mathcal{B}_i]/C_i, V) = \mathrm{Hom}_\Gamma(k[\mathcal{P}], V)$$

for every V . The induced incidence map $k[\mathcal{B}_i]/C_i \rightarrow k[\mathcal{P}]$ is surjective. Its source and target have the same semisimple multiplicities, so it is an isomorphism. Its kernel is $(K_i/C_i) \otimes k$, which is therefore zero. Characteristic-zero scalar extension gives $K_i = C_i$. $\square$

5.2 The exact rank certificate

The primes 409 and 613 are congruent to 1 modulo 204. Thus both fields split Γ . For each of the three designs and each of the eight Galois-orbit representatives, the frozen component matrix has a square minor of order

$$m_{B_i}(V) - m_P(V)$$

whose determinant is nonzero modulo 409. These 24 minors prove the required characteristic-zero ranks. Reconstructing the same 24 ranks modulo 613 is an independent robustness check rather than an additional premise.

The seed stream is round-robin across all 69 two-space orbits. This condition is structural: a lexicographic stream can omit invariant mass, whereas the frozen stream meets every orbit stratum. The certificate stores the orbit/stabilizer input, fixed-space bases, selected row and column indices, determinant residue, and a negative control obtained by removing one essential row or column.

Theorem 5.2 (Minimum trades span the rational kernel). *For $i = 1, 2, 3$,*

$$C_i = K_i.$$

Moreover, the Γ -spans of deterministic prefixes of respectively

$$1\,143, \quad 1\,170, \quad 648$$

minimum-trade seeds already equal K_1, K_2, K_3 .

Proof. The nonzero minors modulo 409 attain the upper bound in theorem 5.1 for one absolute representative in every rational Galois orbit. All circuit vectors are rational, so Galois conjugation transports each rank statement to the other absolute constituents in its orbit. The criterion gives $C_i = K_i$.

For each design, take the largest prefix length used by any of its eight component certificates. Every component has already attained its target rank at that common prefix, giving the three stated bounds. $\square$

Remark 5.3. *The theorem is not a consequence of the census alone. It depends on attaining the exact rank in every irreducible component. Conversely, the minor at 409 is sufficient for the characteristic-zero transfer; the computation at 613 checks the frozen construction independently.*

6 The first toric layer

Fix i and a field k . Let

$$S_i = k[x_B : B \in \mathcal{B}_i]$$

and define the incidence monomial map

$$\pi_i : S_i \longrightarrow k[y_p : p \in \mathcal{P}], \quad x_B \longmapsto \prod_{p \in \text{PG}(B)} y_p.$$

The incidence toric ideal is $I_i = \ker \pi_i$. Since every block contains 15 points, I_i is homogeneous for total degree.

Lemma 6.1 (No linear or quadratic moves). *The ideal I_i has no nonzero binomial of degree one or two.*

Proof. Distinct blocks have distinct point-incidence columns, excluding degree one. Suppose a quadratic relation remains after cancelling common variables, and choose a block B from one side. If that side contains B^2 , its incidence vector has scalar product at least 30 with the column of B . Otherwise it contains B once and the scalar product is at least 15. A different four-space meets B in at most 7 projective points, so the two blocks on the opposite side contribute at most 14. Both cases are impossible. $\square$

Lemma 6.2 (Squarefree cubic classification). *Every nonzero cubic binomial in I_i , after cancellation, is the binomial of a minimum opposite-regulus trade.*

Proof. If a block B is repeated on one side, its scalar product with that side is at least 30. The three distinct blocks on the other side contribute at most $3 \cdot 7 = 21$. Thus both cubic monomials are squarefree. A block shared by the two sides would cancel to a nonzero quadratic, contradicting theorem 6.1. Hence the binomial has three blocks in each leg and total support six. It is a minimum $T_2(1, 4, 8)$ trade, and Krotov's $t = 1$ classification identifies it with an opposite-regulus trade [7]. $\square$

For a monomial M , let

$$\mathcal{F}(M) = \{N : \pi_i(N) = \pi_i(M)\}$$

be its multidegree fiber.

Lemma 6.3 (Two-element fibers). *If $M = x_{B_1}x_{B_2}x_{B_3}$ is one leg of a minimum trade and N is the other leg, then*

$$\mathcal{F}(M) = \{M, N\}.$$

Proof. Every monomial in the fiber has degree three. By theorems 6.1 and 6.2, a second monomial is squarefree, shares no block with M , and forms a minimum regulus trade with it.

The three blocks in M recover

$$Z = B_1 \cap B_2 \cap B_3, \quad U = B_1 + B_2 + B_3.$$

Their images in $\text{PG}(U/Z) \cong \text{PG}(3, 2)$ are three skew lines. Those lines have a unique opposite regulus, which lifts to the three blocks of N . Thus no third monomial occurs. $\square$

Theorem 6.4 (Complete cubic layer). *For each i , the primitive cubics of I_i are exactly the minimum opposite-regulus binomials. Every one is indispensable. Their numbers are*

$$2\,271\,676, \quad 2\,213\,026, \quad 2\,216\,868$$

for $i = 1, 2, 3$, respectively. Equivalently,

$$\beta_{0,3}(I_i) = \beta_{1,3}(S_i/I_i) = |\mathcal{C}_i|.$$

Proof. Theorem 6.2 identifies all cubics, and theorem 6.3 shows that each has a two-element fiber. The indispensable-fiber criterion of Aoki–Takemura–Yoshida implies that the corresponding binomial occurs, up to scalar, in every Markov basis [1]. Since theorem 6.1 excludes lower degrees, these cubics are minimal generators. The exact counts are those of theorem 4.2. $\square$

Remark 6.5 (Higher degrees). *The theorem determines the first nonzero graded layer. It does not prove that I_i is generated by cubics, nor does it determine a complete Markov or Graver basis.*

7 Integral relation lattices

Let

$$L_i = \langle T : T \in \mathcal{C}_i \rangle_{\mathbb{Z}} \subseteq K_i^{\mathbb{Z}}, \quad \Theta_i = K_i^{\mathbb{Z}}/L_i.$$

The rational theorem determines the rank but not the index of L_i . This distinction is the source of the integral phenomenon below.

Lemma 7.1 (Finiteness and local saturation). *The group Θ_i is finite. Let p be a prime. If the image of $L_i \otimes \mathbb{Z}_p$ spans $(K_i^{\mathbb{Z}} \otimes \mathbb{Z}_p)/p(K_i^{\mathbb{Z}} \otimes \mathbb{Z}_p)$, then $\Theta_i[p^\infty] = 0$.*

Proof. The lattice $K_i^{\mathbb{Z}}$ is primitive in $\mathbb{Z}[\mathcal{B}_i]$: its quotient is the image of A_i , a subgroup of the free group $\mathbb{Z}[\mathcal{P}]$. By theorem 5.2, L_i and $K_i^{\mathbb{Z}}$ have the same rational span and hence the same rank. Their quotient is finite.

Put $M = K_i^{\mathbb{Z}} \otimes \mathbb{Z}_p$ and $N = L_i \otimes \mathbb{Z}_p$. The spanning hypothesis is $N + pM = M$. Nakayama's lemma gives $N = M$, so the localized quotient vanishes. $\square$

7.1 The prime 2

Proposition 7.2 (Binary saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[2^\infty] = 0.$$

Proof. In each design, an explicit set of 66 674 minimum trades is linearly independent in $\mathbb{F}_2[\mathcal{B}_i]$. Primitivity of $K_i^{\mathbb{Z}}$ makes $K_i^{\mathbb{Z}}/2K_i^{\mathbb{Z}}$ a 66 674-dimensional subspace of that ambient permutation module. The selected trades lie in this subspace and already have its full dimension. Apply theorem 7.1. $\square$

This statement concerns liftable integral relations, not the complete modular null-design space. The distinction is especially visible at the next prime and is consistent with the modular minimum-support setting of [8].

7.2 The prime 3: the liftable kernel

Let $\alpha : \mathbb{Z}[\mathcal{B}_i] \rightarrow \mathbb{Z}$ be augmentation, $\alpha(\sum n_B B) = \sum n_B$. Summing all point coordinates of an integral incidence relation gives

$$15 \alpha(x) = 0.$$

Thus α vanishes on $K_i^{\mathbb{Z}}$, even though its reduction modulo 3 is not a consequence of the incidence equations.

Lemma 7.3 (Augmentation–Bockstein identification). *For every i ,*

$$K_i^{\mathbb{Z}}/3K_i^{\mathbb{Z}} = \ker_{\mathbb{F}_3} \begin{bmatrix} A_i \\ \alpha \end{bmatrix}.$$

Proof. The locked Smith data have exactly one factor divisible by 3, namely 120, in the 255-row incidence presentation. In particular,

$$\text{rank}_{\mathbb{F}_3} A_i = 254, \quad \text{rank}_{\mathbb{F}_3} \begin{bmatrix} A_i \\ \alpha \end{bmatrix} = 255.$$

The modular kernel of A_i therefore has dimension 66 675, one larger than the reduction of the primitive rank-66 674 integral kernel. The augmented kernel has dimension 66 674 and contains that reduction by the preceding augmentation identity. Equality follows. Equivalently, the missing modular direction is the one-dimensional Tor_1 contribution produced by the factor 120. $\square$

Write $\Gamma = F \times C_3$, where $|F| = 68$, and put

$$R_3 = \mathbb{F}_3[C_3] \cong \mathbb{F}_3[u]/(u^3), \quad u = c - 1.$$

The ideal (u) is the radical of $\mathbb{F}_3\Gamma$, while $\mathbb{F}_3F$ is semisimple. Let $M_i^{(3)}$ denote the liftable kernel in theorem 7.3, and let $\overline{M}_i^{(3)} = M_i^{(3)}/uM_i^{(3)}$ be its semisimple top.

The exact component certificates decompose this top into four Frobenius families, whose Wedderburn dimensions account for all 68 dimensions of $\mathbb{F}_3F$. The images of minimum trades attain total top dimensions

	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$\dim_{\mathbb{F}_3} \overline{M}_i^{(3)}$	22 304	22 338	22 270.

Each component rank is accompanied by a maximal minor and a rank-loss control.

Proposition 7.4 (Ternary saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[3^\infty] = 0.$$

Proof. By theorem 7.3, the module tested by the certificates is exactly the reduction of the integral kernel, not the one-dimension-larger modular kernel. The component ranks say that the minimum trades generate every simple summand of $\overline{M}_i^{(3)}$. Hence

$$M_i^{(3)} = L_i^{(3)} + uM_i^{(3)}.$$

The ideal (u) is nilpotent, so Nakayama's lemma gives $M_i^{(3)} = L_i^{(3)}$. Applying the local criterion to the integral lattices proves the assertion. $\square$

7.3 The prime 17: a free cyclic kernel

The normal subgroup C_{17} acts freely on points and blocks. There are 15 point orbits and 3 937 block orbits. Put

$$R_{17} = \mathbb{F}_{17}[C_{17}] \cong \mathbb{F}_{17}[u]/(u^{17}), \quad u = a - 1.$$

The two permutation modules are free R_{17} -modules of ranks 3 937 and 15. The coinvariant incidence map is surjective. Nakayama therefore makes the full R_{17} -map surjective, and its kernel $M_i^{(17)}$ is free of rank

$$3\,937 - 15 = 3\,922.$$

The quotient $H = \Gamma/C_{17} \cong C_4 \times C_3$ is semisimple in characteristic 17. Its eight simple components have total Wedderburn mass $12 = |H|$. In every component, exact fixed-space matrices show that the minimum trades span

$$M_i^{(17)}/uM_i^{(17)}.$$

Proposition 7.5 (Seventeen-adic saturation). *For $i = 1, 2, 3$,*

$$\Theta_i[17^\infty] = 0.$$

Proof. The eight component ranks generate the complete semisimple top of the free R_{17} -kernel. Thus $M_i^{(17)} = L_i^{(17)} + uM_i^{(17)}$. Nakayama over the local ring R_{17} gives equality, and theorem 7.1 lifts the conclusion to the 17-adic lattices. $\square$

7.4 What the small components exclude

Away from $2 \cdot 3 \cdot 17$, the six rational types of dimensions at most four admit integral component models. For each of the resulting 18 maps, two exact maximal minors have the gcd shown below.

V	$\mathcal{B}_1$	$\mathcal{B}_2$	$\mathcal{B}_3$
$\mathbf{1}$	24	6	24
ε	8	2	128
ι	1024	4096	128
ω	4	216	8
$\varepsilon\omega$	108	12	2
$\iota\omega$	33 554 432	16 777 216	49 152

Every displayed number is supported on $\{2, 3, 17\}$. Bezout combination of the corresponding minors makes each maximal-minor ideal a unit after these primes are inverted.

Proposition 7.6 (Small-component exclusion). *No prime outside $\{2, 3, 17\}$ occurs in any of the 18 component cokernels belonging to*

$$\mathbf{1}, \varepsilon, \iota, \omega, \varepsilon\omega, \iota\omega.$$

This is not a support theorem for Θ_i . The six large maps $\rho/\mathcal{B}_i$ and $\rho\omega/\mathcal{B}_i$ require separate complete-equation-map analysis. A control-prime-selected presentation can display a good-prime rank defect that disappears when the full minimum-circuit space is imposed. Section 8 resolves the selected primes 137, 219 097, and 3 288 036 131; Section 9 then closes the single $\rho/\mathcal{B}_1$ component globally away from 2, 3, and 17.

8 Complete saturation at the three screened good primes

We first record the exact passage from component equations to the integral quotient. The distinction between the complete equation space and a control-prime-selected row set is essential in this section.

Lemma 8.1 (Good-prime component detection). *Let $p \nmid |\Gamma|$, suppose $A_i \otimes \mathbb{F}_p$ is surjective, and let $k/\mathbb{F}_p$ split Γ . Put*

$$Q_i = (K_i^{\mathbb{Z}}/L_i) \otimes k.$$

For every absolutely irreducible $k\Gamma$ -module V , the cokernel of the complete component equation map is dual to $\text{Hom}_{k\Gamma}(Q_i, V)$. Consequently, if every complete component map has its structural target rank, then $\Theta_i \otimes \mathbb{F}_p = 0$, and hence $\Theta_i[p^\infty] = 0$.

Proof. Reduction gives an exact sequence

$$0 \longrightarrow Q_i \longrightarrow k[\mathcal{B}_i]/(L_i \otimes k) \longrightarrow k[\mathcal{P}] \longrightarrow 0.$$

Maschke's theorem makes $k\Gamma$ semisimple. Applying $\text{Hom}_{k\Gamma}(-, V)$ identifies the indicated cokernel with the dual of $\text{Hom}_{k\Gamma}(Q_i, V)$. The word “complete” is necessary: replacing all minimum-circuit equations by a row subset selected at a different prime need not preserve rank after reduction modulo p . If every absolute type vanishes after a splitting extension, faithful scalar extension gives $Q_i = 0$. Finally, finiteness of Θ_i and Nakayama's lemma give $\Theta_i[p^\infty] = 0$. $\square$

At $p = 137$, the hypotheses hold: $137 \nmid 204$, every projector denominator is a unit, and the incidence Smith form has no 137-factor.

8.1 Why the selected-row obstruction is not a quotient class

For $\rho/\mathcal{B}_1$, a 1312-row presentation selected for full rank at the control prime 409 has rank 1311 at

$$\mathfrak{p} = (137, \zeta_{17} - 122).$$

Its certified residual has rank 256 rather than 257, and its local selected-presentation cokernel has length one. Those calculations are exact, including the Hensel lift modulo 137^2 , but they concern the selected presentation only.

They do not prove a class in Θ_1 : additional minimum-circuit equations, absent from the 409-selected set, may restore the structural target rank at 137. This is what happens.

8.2 The full-circuit restoration

There are 11274 Γ -orbits of internal minimum circuits in $\mathcal{B}_1$. One seed per orbit suffices by equivariance. At $\zeta_{17} = 122$, the absolute ρ -representation has dimension 4; the block multiplicity space has dimension 1317, while the point-induced kernel has dimension 5. Hence the maximum circuit rank is

$$1317 - 5 = 1312.$$

Deterministic sparse elimination over $\mathbb{F}_{137}$, in lexicographic orbit and coordinate order, gives

processed circuit orbits	rank
500	1260
1000	1309
2000	1311
2864	1312.

The restoring source row is coordinate 0 of orbit 2864 (zero-based identifier (2863, 0)). An independent reconstruction forms the selected 1312×1312 minor and obtains

$$\det = 56 \pmod{137}.$$

Removing the restoring row leaves rank 1311, and duplicating a pivot column makes the determinant zero.

Theorem 8.2 (Complete $\rho/\mathcal{B}_1$ contribution above 137). *The complete minimum-circuit component map for $\rho/\mathcal{B}_1$ is full at every prime above 137. In particular,*

$$\text{coker}(F_{\rho,1}^{\text{full}})[137^\infty] = 0.$$

The length-one cokernel at $(137, \zeta_{17} - 122)$ belongs only to the control-prime-selected presentation and is not a quotient of Θ_1 .

Proof. At fifteen of the sixteen split primes, the earlier selected row set already contains a nonzero maximal minor, so the complete map is full there. At the remaining root 122, the full-circuit minor above has nonzero determinant. Thus every localization above 137 is full. $\square$

8.3 The other five large maps

For ρ , the sixteen primes above 137 are the degree-one roots of Φ_{17} . For $\rho\omega$, put

$$k_2 = \mathbb{F}_{137}[y]/(y^2 + y + 1).$$

If r runs over the sixteen order-17 roots in $\mathbb{F}_{137}$, then

$$\alpha_r = r^6 y^2, \quad \alpha_r^3 = r, \quad \alpha_r^{17} = y$$

has order 51, and the Frobenius pairs $\{\alpha_r, \alpha_r^{137}\}$ represent the sixteen degree-two primes of $\mathbb{Z}[\zeta_{51}]$ above 137.

The exact original-column presentations give:

map	rows	columns	residue degrees	full maximal minors
$\rho\omega/\mathcal{B}_1$	1305	10480	2^{16}	16/16
$\rho/\mathcal{B}_2$	1314	5276	1^{16}	16/16
$\rho\omega/\mathcal{B}_2$	1304	10472	2^{16}	16/16
$\rho/\mathcal{B}_3$	1310	5260	1^{16}	16/16
$\rho\omega/\mathcal{B}_3$	1306	10488	2^{16}	16/16.

For the quadratic rows, multiplication by $a + by$ is expanded as

$$\begin{bmatrix} a & -b \\ b & a - b \end{bmatrix}.$$

A second parser uses the Frobenius-conjugate root, transposes every matrix, reproduces all 80 determinant residues, and kills each determinant after duplicating one field column.

Theorem 8.3 (Complete 137-primary quotients). *For the three frozen configurations,*

$$\Theta_1[137^\infty] = \Theta_2[137^\infty] = \Theta_3[137^\infty] = 0.$$

Proof. Theorem 7.6 excludes 137 from all eighteen small component cokernels. Theorem 8.2 closes the formerly exceptional $\rho/\mathcal{B}_1$ component, and the eighty nonzero maximal minors close the other five large maps. The complete-component criterion in theorem 8.1 gives $\Theta_i \otimes \mathbb{F}_{137} = 0$ for every i ; finiteness then excludes all 137-primary torsion. $\square$

Remark 8.4 (Exact boundary). *The theorem is complete at the rational prime 137. It does not classify another rational prime or determine the global Smith invariant factors of the finite groups Θ_i . In particular, singular minors selected at 409 above other candidate primes must first be audited against the complete circuit equation space.*

8.4 A second complete prime envelope

The next selected-presentation signal occurs at $p = 219\,097$. Since

$$219\,097 - 1 = 2^3 \cdot 3^2 \cdot 17 \cdot 179, \quad 219\,097 \equiv 1 \pmod{51},$$

both cyclotomic orders split completely. Thus a ρ -map has sixteen degree-one localizations and a $\rho\omega$ -map has thirty-two.

For $\rho/\mathcal{B}_1$, the complete circuit map restores rank 1 312 at the historical root 91085, again at source row (2863, 0). The independent maximal minor has determinant

$$213269 \pmod{219097},$$

and the prefix without the restoring row has rank 1 311.

For the other five maps, exact original-column maximal minors give

map	target rank	localizations	nonzero minors
$\rho\omega/\mathcal{B}_1$	1305	32	32
$\rho/\mathcal{B}_2$	1314	16	16
$\rho\omega/\mathcal{B}_2$	1304	32	32
$\rho/\mathcal{B}_3$	1310	16	16
$\rho\omega/\mathcal{B}_3$	1306	32	32.

An independent parser enumerates the roots from a primitive generator, reconstructs the transpose of every selected matrix, reproduces all 128 determinants, and obtains determinant zero in all 128 duplicate-column controls.

Theorem 8.5 (Complete 219 097-primary quotients). *For the three frozen configurations,*

$$\Theta_1[219097^\infty] = \Theta_2[219097^\infty] = \Theta_3[219097^\infty] = 0.$$

Proof. Theorem 7.6 excludes 219 097 from all eighteen small component cokernels. The full-circuit determinant closes $\rho/\mathcal{B}_1$, and the 128 nonzero maximal minors close the other five large maps. The good-prime complete-component criterion theorem 8.1 gives $\Theta_i \otimes \mathbb{F}_{219097} = 0$ for all three designs; finiteness excludes 219 097-primary torsion. $\square$

Remark 8.6 (Remaining integral boundary). *The theorem removes 219 097 completely from the possible support. It does not classify another rational prime or the global Smith invariant factors.*

8.5 A third complete prime envelope

The last selected-presentation signal is $p = 3\,288\,036\,131$. Here

$$p - 1 = 2 \cdot 5 \cdot 17 \cdot 2243 \cdot 8623, \quad p \equiv 1 \pmod{17}, \quad p \equiv 2 \pmod{3}.$$

Thus Φ_{17} has sixteen linear factors, while Φ_{51} has sixteen irreducible quadratic factors.

For $\rho/\mathcal{B}_1$, the complete circuit map restores rank 1 312 at the historical root 1 120 906 769, again at source row (2863, 0). The independently reconstructed maximal minor has

$$\det = 1\,471\,654\,912 \pmod{3\,288\,036\,131},$$

and the prefix without the restoring row has rank 1 311.

For the $\rho\omega$ -maps, use

$$k_2 = \mathbb{F}_p[y]/(y^2 + y + 1), \quad \alpha_r = r^6 y^2,$$

where r runs through the order-17 roots. Then $\alpha_r^3 = r$, $\alpha_r^{17} = y$, and the Frobenius pairs give the sixteen quadratic primes. The five remaining maps have the exact screen

map	target rank	residue degree	localizations	nonzero minors
$\rho\omega/\mathcal{B}_1$	1305	2	16	16
$\rho/\mathcal{B}_2$	1314	1	16	16
$\rho\omega/\mathcal{B}_2$	1304	2	16	16
$\rho/\mathcal{B}_3$	1310	1	16	16
$\rho\omega/\mathcal{B}_3$	1306	2	16	16.

The independent route finds the least primitive generator 2, uses the Frobenius-conjugate quadratic representatives, transposes every matrix, reproduces all 80 determinant residues, and obtains determinant zero in all 80 duplicate-field-column controls.

Theorem 8.7 (Complete 3 288 036 131-primary quotients). *For the three frozen configurations,*

$$\Theta_1[3288036131^\infty] = \Theta_2[3288036131^\infty] = \Theta_3[3288036131^\infty] = 0.$$

Proof. Theorem 7.6 excludes 3 288 036 131 from all eighteen small component cokernels. The full-circuit determinant closes $\rho/\mathcal{B}_1$, and the 80 nonzero maximal minors close the other five large maps. The complete-component criterion in theorem 8.1 gives $\Theta_i \otimes \mathbb{F}_p = 0$; finiteness excludes all p -primary torsion. $\square$

Remark 8.8 (Selected signals versus global support). *All three selected good-prime rank losses, at 137, 219 097, and 3 288 036 131, disappear for the complete maps. This finite list alone does not prove that no untested rational prime divides a global Smith factor. The next section supplies that global exclusion away from 2, 3, 17 for the single complete $\rho/\mathcal{B}_1$ component. The other five large maps, the support of the complete Θ_i , and their global invariant factors remain open.*

9 A global localized theorem for the $\rho/\mathcal{B}_1$ component

The three finite-prime restorations in Section 8 show that the rank losses inherited from the control-prime presentation are not intrinsic at those primes. They do not, by themselves, exclude all other primes. We now replace that finite screen by an exact characteristic-zero certificate for the same complete component map.

Put

$$R = \mathbb{Z}[\zeta], \quad \zeta = \zeta_{17}, \quad R_S = R \left[\frac{1}{2 \cdot 3 \cdot 17} \right].$$

The power basis

$$1, \zeta, \dots, \zeta^{15}$$

is an integral basis of the cyclotomic ring R [13]. In the fixed bases used throughout the component audit, the complete minimum-circuit map is

$$A = F_{\rho,1}^{\text{full}} : R^{1317} \longrightarrow R^{1312}.$$

Thus its cokernel measures only the residual $\rho/\mathcal{B}_1$ component obstruction. It is not the complete group Θ_1 .

9.1 Exact lift and determinant reconstruction

Lemma 9.1 (Common cyclotomic lift). *The 1317 source columns and every complete minimum-circuit equation for A have one exact realization over R in the power basis above. Under a residue homomorphism sending ζ to an element of order 17, this realization reduces entrywise to the corresponding complete finite component map.*

Proof. The fixed-space columns are first normalized as exponent vectors modulo 17. Every representation entry is then a signed sum of those powers, reduced by

$$1 + \zeta + \dots + \zeta^{16} = 0.$$

Circuit rows are formed from the same six signed block columns before any specialization. Consequently evaluation at a seventeenth root commutes with both column construction and row formation. The independent replay checks all 1317 normalized columns and the entrywise reductions used at the three screened primes; no independently chosen finite-field basis is lifted back to characteristic zero. $\square$

For completeness, we record why the determinant lifts below are exact rather than heuristic modular reconstructions.

Lemma 9.2 (Certified signed CRT reconstruction). *Let B be either square matrix in theorem 9.3, and write*

$$\det B = \sum_{j=0}^{15} c_j \zeta^j.$$

For each complex embedding, Hadamard's inequality bounds the determinant by the square root of the product of the certified row energies. Fourier inversion in the seventeenth cyclotomic basis then gives

$$\max_j |c_j| \leq \frac{32}{17} \max_{\sigma: R \rightarrow \mathbb{C}} |\sigma(\det B)|.$$

If the product M of the split CRT primes is larger than twice this coefficient bound, the sixteen signed coefficients c_j are uniquely determined by their residues modulo M .

Proof. Hadamard gives $|\det \sigma(B)|^2 \leq \prod_r E_r$ from the row-energy bounds E_r . Applying the inverse seventeenth-root transform and eliminating the coefficient of ζ^{16} with $\Phi_{17}(\zeta) = 0$ gives the displayed $32/17$ bound. Finally, two integers in the signed interval $(-M/2, M/2)$ with the same residue modulo M are equal. Apply this coordinatewise. $\square$

Proposition 9.3 (Two exact maximal minors). *The matrix A has two explicitly displayed 1312×1312 minors with determinants $\delta_G, \delta_R \in R$. Their independent exact reconstructions have the following certified data.*

<i>minor</i>	<i>entries</i>	<i>check</i>	<i>CRT primes</i>	<i>bound bits</i>	<i>modulus bits</i>	<i>norm bits</i>
δ_G	7830	89 mod 103	96	2937	2967	4833
δ_R	7819	107 mod 137	93	2872	2874	2003

The largest reconstructed coefficient has respectively 304 and 128 bits.

Proof. For δ_G , the complete map at $(103, \zeta - 8)$ has rank 1312. Replacing the unique deficient column in the original selected minor gives determinant 89 and an exact sparse lift with 7830 nonzero entries.

For δ_R , scan the 11 274 circuit orbits in reverse canonical order and, inside an orbit, scan the four representation coordinates in reverse order. The first full row basis is reached after 11 069 orbits and 44 276 rows. With canonical columns $0, \dots, 1311$, its determinant is 107 at $(137, \zeta - 122)$, and its exact lift has 7819 nonzero entries.

In both cases an independent binary parser reconstructs the transpose, checks the finite determinant, and obtains rank 1311 after the designated deletion or duplication mutation. The split-prime congruences, row-energy products, and inequalities in theorem 9.2 are then replayed for every CRT prime. Since the recorded moduli exceed twice the proved bounds, the lifted coefficient vectors, and hence their exact cyclotomic norms, are unique. $\square$

9.2 The two-minor ideal certificate

We use two standard consequences of Fitting ideals and integral Smith theory [11, 12, 14].

Lemma 9.4 (Two-minor Fitting criterion). *Let T be a commutative ring and $f : T^n \rightarrow T^m$, with $n \geq m$. If two maximal minors α, β of f satisfy $(\alpha, \beta)T' = T'$ after a flat base change $T \rightarrow T'$, then*

$$\text{coker}(f) \otimes_T T' = 0.$$

Proof. The zeroth Fitting ideal of $\text{coker}(f)$ is generated by all maximal minors of a presentation matrix and commutes with base change. It therefore contains $(\alpha, \beta)T' = T'$. Thus its localization at every prime of T' is the unit ideal. Some maximal minor is a unit in each local ring, so the localized map is surjective. The finitely generated cokernel has empty support and is zero. $\square$

Lemma 9.5 (Multiplication-lattice certificate). *For $\alpha \in R$, let $M(\alpha) \in \text{Mat}_{16}(\mathbb{Z})$ be multiplication by α in the integral power basis. Then*

$$\text{im}_{\mathbb{Z}} M(\alpha) = \alpha R.$$

Consequently the columns of

$$[M(\alpha) \ M(\beta)] \in \text{Mat}_{16 \times 32}(\mathbb{Z})$$

span $\alpha R + \beta R$, and the nonzero Smith entries of this integer matrix are the additive invariant factors of $R/(\alpha R + \beta R)$.

Proof. The columns of $M(\alpha)$ are $\alpha, \alpha\zeta, \dots, \alpha\zeta^{15}$ in coordinates, hence span exactly the principal ideal αR . Concatenating two such matrices adds their column lattices. Smith normal form computes the quotient of the ambient free abelian group by that full-rank sublattice. $\square$

Theorem 9.6 (Global localized saturation of the $\rho/\mathcal{B}_1$ component). *For the complete minimum-circuit equation map*

$$A = F_{\rho,1}^{\text{full}} : R^{1317} \longrightarrow R^{1312},$$

one has

$$\text{coker}(A) \otimes_R R_S = 0.$$

Equivalently, $A \otimes_R R_S$ is surjective. Thus the complete $\rho/\mathcal{B}_1$ component cokernel has no prime-ideal support away from the rational primes 2, 3, and 17.

Proof. Apply theorem 9.5 to the two determinants in theorem 9.3. The independently replayed Smith form of

$$[M(\delta_G) \ M(\delta_R)]$$

has diagonal

$$\underbrace{17, \dots, 17}_{11 \text{ times}}, \quad \underbrace{289, \dots, 289}_{5 \text{ times}}.$$

Therefore

$$[R : \delta_G R + \delta_R R] = 17^{11}(17^2)^5 = 17^{21} = 69\,091\,933\,913\,008\,732\,880\,827\,217.$$

After inverting 17, and hence over R_S , the two determinants generate the unit ideal. Both are maximal minors of A , so theorem 9.4 proves the assertion. $\square$

Remark 9.7 (Why a stable determinant factor is not support). *Neither displayed determinant is an S -unit by itself. Moreover, several one-column replacements and a reverse-column basis built on the same frozen row set retain a large common non- S factor. That persistence is a presentation artifact: the genuinely different reverse-row minor removes it from the ideal sum.*

As a negative control, replacing δ_G by the original $G0$ determinant leaves, after removal of 2, 3, and 17, the nonunit residual

$$47\,821\,880\,003\,927\,349\,029.$$

Thus neither a factor of one norm nor its survival across a closely related minor family can be promoted to support of the complete cokernel.

Remark 9.8 (Exact boundary). *Theorem 9.6 closes one of the six large component maps. The global gates for*

$$\rho\omega/\mathcal{B}_1, \quad \rho/\mathcal{B}_2, \quad \rho\omega/\mathcal{B}_2, \quad \rho/\mathcal{B}_3, \quad \rho\omega/\mathcal{B}_3$$

remain open. In particular, the theorem does not prove $\text{Supp}(\Theta_i) \subseteq \{2, 3, 17\}$, does not determine the global Smith groups, and is not a tomography or Markov-connectivity statement.

10 Discussion, prior art, and open directions

10.1 The rational–integral contrast

The main result has a deliberately asymmetric scope. The minimum regulus trades generate every rational relation in all three configurations. At the integral level we prove saturation at 2, 3, 17, 137, 219 097, 3 288 036 131. Globally, the exact two-minor certificate closes one of the six large maps, namely the complete $\rho/\mathcal{B}_1$ component, away from 2, 3, 17. Equality of the full trade and relation lattices remains open because the other five large component gates have not been computed. Thus

$$\text{minimum local geometry} \implies \text{full rational generation} \not\Rightarrow \text{global integral saturation}.$$

The character table also shows that common incidence parameters and a common group do not make the relation modules isomorphic.

The general theories used here have distinct ownership. Cho’s characteristic-zero null-design decomposition and minimum subspace null-design work provide the complete-Grassmann background [2, 3]. Krotov supplies the minimum-support theorem and the $t = 1$ classification [7]. The ordinary-subset orbit-spanning result of Ghorbani–Kamali–Khosrovshahi is the closest structural analogue of theorem 5.2 [4]. It uses the full symmetric action on subsets, whereas the present theorem concerns selected binary subspace constituents and the order-204 group.

For toric ideals, Aoki–Takemura–Yoshida own the indispensable-fiber criterion [1]. Betti–Grate–Holleben–Salizzoni develop the current incidence-toric/null-design language for ordinary subset incidence matrices [10]. The present finite calculation is the proof that the cubic fibers of these selected Grassmann configurations have exactly the two regulus monomials. Classical integral-design modules and circuit lattices are represented by [6, 5]; the component computation in Section 8 is specific to the locked triple. These comparisons are scope boundaries, not priority claims.

Fitting ideals, cyclotomic arithmetic, CRT reconstruction, and Smith normal form are standard tools [11, 12, 13, 14]. The contribution here is their exact application to the frozen complete component map and the independently replayed two-minor certificate. No method-firstness or priority claim is made.

10.2 Tomography and the P42 interface

There is a useful but logically separate sensing result. The 97 155 three-dimensional subspaces of $\mathbb{F}_2^8$ split into 515 Γ -orbits, with orbit-size profile

$$51^5, \quad 102^{70}, \quad 204^{440}.$$

The selected three-space versus four-space incidence map

$$W_{3,4} : \mathbb{Q}[\mathcal{B}_i] \longrightarrow \mathbb{Q}[\mathrm{Gr}_2(8, 3)]$$

has column rank 66 929, so it is injective on the entire block module and, in particular, on every K_i . It gives a lossless rational sensor system for the relations.

This statement is not used in theorems 5.2, 8.2, 8.3, 8.5, 8.7 and 9.6. The temporary selected-row rank losses at 137, 219 097, and 3 288 036 131 were presentation phenomena, not tomography failures and not integral quotient classes. Conversely, no tomography statement is used in the exact cyclotomic or Fitting argument of theorem 9.6. The broader comparison of sensor families, minimal sensor sets, and reconstruction belongs to the planned P42 tomography paper. The synchronized interface is:

P39	P42
relation modules and circuit generators	measurement maps and reconstruction
one lossless $W_{3,4}$ check	systematic sensor comparison
integral component audit	distinguish sensing loss from lifting loss.

10.3 Remaining global integral gates

At 137, 219 097, and 3 288 036 131, all six large maps are classified and theorems 8.3, 8.5 and 8.7 gives complete three-design vanishing. These were exactly the good-prime signals found by the frozen selected-row search. Their disappearance under complete-map audits shows that the candidate list was not a support theorem.

Theorem 9.6 goes beyond that finite list for one map: every prime ideal outside 2, 3, 17 is excluded from the complete $\rho/\mathcal{B}_1$ cokernel. The remaining integral tasks are therefore:

- compute an exact localized maximal-minor ideal certificate for each of the other five large component maps;
- combine those results with theorem 7.6 to prove or refute $\mathrm{Supp}(\Theta_i) \subseteq \{2, 3, 17\}$;
- compute the exact global Smith invariant factors, including the remaining 2-, 3-, and 17-primary structure.

A further finite-prime search cannot close these questions. The successful $\rho/\mathcal{B}_1$ route shows the required form of a certificate: one common integral cyclotomic basis, exact characteristic-zero determinants from genuinely different row bases, and their ideal sum rather than a rational norm gcd. The failure of closely related same-row minors in theorem 9.7 is the reason row-basis diversity matters. The five repetitions are deliberately deferred; none is needed for the rational, cubic, selected-prime, or one-component global theorems proved here.

10.4 Geometric and family questions

Krotov’s bound gives support 30 and leg volume 15 for $T_2(2, 4, 8)$, but the $t = 1$ uniqueness theorem used in this paper does not classify that case. An exact next question is a two-colour census: one leg in $\mathcal{B}_i$, the other in $\mathcal{B}_j$. The canonical construction can be searched orbitally, but exhaustiveness requires a separate classification theorem.

A second direction is to replace the locked finite triple by a general selected q -subspace configuration. The missing theorem is not the minimum-trade bound; it is nonzero projection of the selected minimum trades to every null-design constituent. The component-rank criterion in theorem 5.1 isolates exactly what such a theorem would have to prove.

Finally, the higher-degree parts of I_i remain open. Cubic indispensability neither implies cubic generation nor determines the complete Markov or Graver basis. The P39 and P42 programs meet naturally at the sequence

$$\mathbb{Q}\text{-generation} \longrightarrow \mathbb{Z}\text{-lifting} \longrightarrow \text{Markov connectivity},$$

but the three arrows should remain separate until their respective integral and higher-degree obstructions are classified.

A Exact certificate contracts

The abstract arguments in the paper reduce the locked finite statements to exact character, rank, determinant, and enumeration inputs. This appendix specifies what must be retained for those inputs to be independently checkable.

A.1 Characters and relation modules

The representation certificate contains:

1. the 24 conjugacy-class representatives and class sizes;
2. the action convention for a, b, c and a check of the group laws;
3. fixed-point counts on $\mathcal{P}, \mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$;
4. the absolute character table, Galois grouping, and rational type labels;
5. exact inner products giving the point, block, and kernel multiplicities;
6. dimension and class-function reconstruction checks.

Changing a class label, type ordering, or action convention must invalidate the reconstruction rather than silently permute a table.

A.2 Minimum-trade census

The census contract binds the digests and canonical orderings of the three block sets. Every accepted trade records Z, U , the two ordered reguli, the six lifted blocks, and the canonical unoriented sign convention. Verification checks:

1. all six blocks belong to the selected constituent;
2. the 255 point coordinates balance;

3. the restricted rank, five-column deletions, and active Smith form;
4. canonical deduplication and complete Γ -orbit closure;
5. agreement of template, membership, and orbit-counting passes.

Exhaustiveness is the combination of Krotov's $t = 1$ classification and complete enumeration of the corresponding (Z, U) -templates.

A.3 Rational component ranks

For each pair (i, V) , the full-kernel certificate stores:

1. block and point orbit representatives with stabilizers;
2. exact bases of V^H and the Frobenius-reciprocity coordinate convention;
3. the round-robin seed stream across all 69 two-space orbits;
4. the dimensions $m_{B_i}(V), m_P(V)$ and target rank;
5. row and column indices of a target-order square minor;
6. its nonzero determinant modulo 409;
7. an independent rank/minor reconstruction modulo 613;
8. a deletion or duplication control that loses the certified rank.

There are $3 \cdot 8 \cdot 2 = 48$ prime/component rank records. The 409 records are the characteristic-zero witnesses; the 613 records test construction stability.

A.4 Local saturation

At 2, the witness lists 66 674 explicit circuits per design. An independent replay reconstructs each point balance and obtains full rank with the opposite pivot convention.

At 3, the contract first verifies the Smith/Bockstein interface:

$$\text{rank } A_i = 254, \quad \text{rank}[A_i; \alpha] = 255,$$

together with the unique 3-divisible Smith factor 120. It then checks the F -representation laws, Wedderburn mass 68, all four component-family top ranks, and rank-loss mutations.

At 17, verification includes free C_{17} -actions, orbit counts 15 and 3 937, surjectivity on coinvariants, free kernel rank 3 922, the eight $C_4 \times C_3$ tops, Wedderburn mass 12, and one rank-loss control per component.

For the 18 small good-prime maps, the matrices and two selected maximal minors are reconstructed over the stated integral orders. Exact determinants and a Bezout gcd supported on $\{2, 3, 17\}$ certify the localized unit ideal.

A.5 The 137-component

The historical selected-presentation contract for $\rho/\mathcal{B}_1$ retains:

1. the exact $1\,312 \times 5\,268$ wide component stream;
2. all 1\,055 pivots, their unit classes, and transition hashes;
3. the byte-bound $257 \times 3\,373$ residual;
4. direct enumeration of the sixteen roots of Φ_{17} ;
5. fifteen nonzero 257-minors and the deficient 256-minor;
6. the root-122 residual left-kernel vector;
7. the Hensel lift and the 256-block, Schur scalar, and 257-minor modulo 137^2 ;
8. the transported 1\,312-coordinate covector and its pairing with coordinate 1\,281;
9. duplicate-column, coordinate, root-label, and determinant mutations that the verifier rejects.

These receipts certify a length-one cokernel of that selected presentation, not of the complete circuit map.

The complete-map correction additionally retains:

1. the deterministic list of all 11\,274 minimum-circuit orbit representatives in $\mathcal{B}_1$;
2. the dimension contract $1317 - 5 = 1312$;
3. the first restoring row $(2863, 0)$;
4. 1\,312 selected full-circuit rows and pivot columns;
5. an independently reconstructed determinant $56 \bmod 137$;
6. prefix-rank 1311, duplicate-column kill, root-order check, and identical normal/optimized verifier receipts.

For the remaining five large maps, the contract additionally retains:

1. one full-row-rank 409 receipt and last-row kill per map;
2. all primitive original columns and the entrywise exact-wide replay;
3. sixteen order-17 roots and, for $\rho\omega$, the sixteen Frobenius orbits in $\mathbb{F}_{137}[y]/(y^2 + y + 1)$;
4. eighty nonzero maximal-minor determinants in total;
5. an independent conjugate-root/transposed reconstruction;
6. eighty duplicate-field-column determinant kills;
7. a deterministic closeout for these five maps, combined in the corrected theorem with the eighteen small components and the full-circuit $\rho/\mathcal{B}_1$ certificate.

No modulo- 137^2 receipt is required for these five maps because every localization already contains a unit maximal minor.

A.6 The 219 097-component

The complete-envelope contract at $p = 219\,097$ retains:

1. the complete $\rho/\mathcal{B}_1$ circuit stream at the historical root 91 085, with first restoring row (2863, 0);
2. a selected full-circuit maximal minor with determinant 213 269 mod 219 097, together with prefix rank 1311;
3. the target ranks and the localization counts 32, 16, 32, 16, 32 for the other five large maps;
4. all 128 nonzero original-column determinant residues and root labels, with the root sets regenerated from the least primitive generator 5;
5. an independent exact-wide parser and transposed reconstruction of every selected matrix;
6. all 128 duplicate-column determinant kills, root-label mutation rejections, and byte-identical normal/optimized receipts;
7. the combination with the eighteen small-component gcd exclusions used to obtain the complete primary triple.

No modulo-219 097² receipt or local-length computation is required: every complete localization already contains a unit maximal minor.

A.7 The 3 288 036 131-component

The complete-envelope contract at $p = 3\,288\,036\,131$ retains:

1. the complete $\rho/\mathcal{B}_1$ circuit stream at root 1 120 906 769, with first restoring row (2863, 0);
2. a 1 312-square full-circuit minor with determinant 1 471 654 912 mod 3 288 036 131, prefix rank 1311, and duplicate-column kill;
3. sixteen degree-one Φ_{17} primes and sixteen degree-two Φ_{51} primes in $\mathbb{F}_p[y]/(y^2 + y + 1)$;
4. the target ranks and the sixteen localizations for each of the other five large maps, eighty in total;
5. all eighty nonzero original-column determinant residues;
6. independent primitive-root enumeration, Frobenius-conjugate evaluations, transposed scalar or doubled matrices, and eighty duplicate-field-column kills;
7. scalar-root and quadratic-orbit mutation rejections, complete input hashes, and byte-identical normal/optimized receipts;
8. the combination with the eighteen small-component gcd exclusions used in theorem 8.7.

No modulo- p^2 or local-length receipt is required because every complete localization contains a unit maximal minor.

A.8 Global $\rho/\mathcal{B}_1$ saturation

The characteristic-zero contract for theorem 9.6 retains:

1. the common $\mathbb{Z}[\zeta_{17}]$ exponent-basis lift of all 1 317 source columns and the entrywise finite-reduction replay;
2. the exact G1R replacement minor and manifest, with SHA-256

```
007D34CA9EBAB7268CA1CB55A1E81BF3E83B7983EE5FB8DD4300BEB41E13B155
B892C0A7BA9419A8C475FFACF05E63149E5A7F11AEBF443D0F01E96CCB0C96DF
```

together with the independently reconstructed determinant receipt;

3. the reverse-row selection order, its 11 069-orbit/ 44 276-row stopping point, and exact binary and manifest hashes

```
C3F857FDEAE82E8680B3EFB0AA281B59661E04E1E3E7C6D689A28A3FF7F45439
B2D696537A73C30E91D31A8FB3957CEF8F5826ACC8EED66E6881A38AB10EA82B
```

4. every split CRT prime and determinant residue, the exact row-energy histograms, signed coefficient bounds, and modulus inequalities for both determinants;
5. independent cyclotomic multiplication, all sixteen conjugates, exact norms, diagnostic finite determinants, coefficient mutations, prefix-rank loss, and duplicate-column kills;
6. the two 16×16 multiplication matrices and the Smith form of their 16×32 concatenation, with diagonal $17^{11}, 289^5$ and index 17^{21} ;
7. the G0/reverse-row negative pair, whose ideal-sum index retains the verified non- S residual 47 821 880 003 927 349 029;
8. byte-identical normal and optimized executions of the final fail-closed verifier.

The decisive ideal-sum artifact is

```
results/s449f_rho_d1_g1_reverse_row_ideal_sum.json
SHA-256 5CB3D78D948E53F3A38B4E85DC9F9C6F9E195839D7B432C544750967B9EFF078
```

and the independent final receipt is

```
results/s449f_rho_d1_global_gate_verification.json
SHA-256 511CD53641E7426650FAA3799BA7D7F3B514F51CA98B69766ADB5E00AE33EBDD
```

with status `PASS_RHO_D1_GLOBAL_GATE_AWAY_FROM_S`. These artifacts close one component only. They do not certify any of the other five large maps or a complete Θ_i .

B Reproducibility and source boundary

The companion package is organized relative to its repository root. The principal fail-closed replay of the characteristic-zero gate is

```
python scripts/verify_all.py
python -O scripts/verify_all.py
python scripts/check_manifest.py
```

It runs four independent verifiers:

```
scripts/s449c_verify_g1r_replacement_minor.py
scripts/s449c_verify_g1r_principal_determinant.py
scripts/s449f_verify_reverse_row_determinant.py
scripts/s449f_verify_rho_d1_global_gate.py
```

The expected SHA-256 receipts, in the same order, are

```
95D0D02E6D233DD25CEA69C07C33A3BCF2D850DE3708E2A1115AB6FEDFE56A15
0077E7FCB73D61EF584E3E503CDA539743A118AA631261A0C2DD75BFC8F1DD84
B7F420235A9947C12E28CD5A887EF883BD7F8D0F37836864B18C6365416BDE88
511CD53641E7426650FAA3799BA7D7F3B514F51CA98B69766ADB5E00AE33EBDD
```

The final verifier independently reconstructs the cyclotomic products and norms, both multiplication lattices, the 16×32 Smith form, and the G0/reverse-row nonunit negative control. Its conclusion is restricted to the complete $\rho/\mathcal{B}_1$ component over $\mathbb{Z}[\zeta_{17}, 1/(2 \cdot 3 \cdot 17)]$. Generated JSON is canonical, and the normal and optimized runs are required to be byte-identical.

Evidence tiers. The package includes the exact binary minors, CRT inputs and raw outputs needed for a clean replay of the new global gate. It also includes frozen compact receipts and the complete minimum-circuit manifest for the earlier census, rational-generation, local-saturation, and selected-prime theorem blocks. The much larger historical raw matrix workspace remains in the source-locked P39 research archive and is not duplicated in the reviewer-sized package. Thus the new global theorem has a self-contained deep replay, while the earlier computer-assisted blocks are distributed with a hash-bound audit trail. This distinction is recorded in the companion reviewer guide and claim ledger.

Evidence levels. Krotov’s classification and the indispensable-fiber criterion are imported theorems. The local circuit proof, semisimple criteria, Bockstein identification, Nakayama reductions, good-prime lemma, CRT uniqueness, multiplication-lattice interpretation, and Fitting implication are mathematical arguments in this paper. The character tables, exhaustive censuses, component ranks, exact minors, determinant coefficients, Smith diagonal, and covectors are exact finite inputs with independent replays or frozen hash-bound receipts. Resource stops from earlier wide Smith eliminations are not used as evidence.

Source boundary. The prior-art search is dated and bounded. It found no source giving the same rational-spanning, cubic-fiber, or complete primary statements at 137, 219 097, and 3 288 036 131 for this locked triple, but that negative result is not a novelty or priority certificate. The paper therefore makes no “first” claim and does not generalize from the three selected configurations without a separate family argument. General Fitting, Smith, CRT, and cyclotomic tools are standard; no method-priority claim is made for their use in theorem 9.6.

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